

Lukas Kugler

Curriculum Vitae

Postdoc Scientist, University of Ljubljana
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PUBLICATIONS

Kugler, L., Serafin, S., Hochebner, A., Pierotti, N., & Weissmann, M. (2026). The benefit of neighbourhood methods for the assimilation of cloud-affected radiances evaluated in observing-system simulation experiments. Accepted for publication in QJRM.

El Gharamti, M., Kershaw, H., Raeder, K., Raczka, B., Johnson, B., Smith, M., Anderson, J. L., Amrhein, D., Collins, N., Grooms, I., & **Kugler, L.** (2025). The Data Assimilation Research Testbed: A Robust, Scalable Software Facility with Groundbreaking Capabilities for Model-Data Integration. BAMS, 106(11), E2328-E2345. <https://doi.org/10.1175/BAMS-D-24-0214.1>

Kugler, L., & Weissmann, M. (2025). Effects of observation operator nonlinearity on the assimilation of visible and infrared radiances in ensemble data assimilation. QJRM, 151(770), 1-14. <https://doi.org/10.1002/qj.4970>

Kugler, L., & Weissmann, M. (2025). The synergy of assimilating visible and infrared radiances and radar observations. QJRM, 149(757), 3623-3644. <https://doi.org/10.1002/qj.4938>

Necker, T., Wolfgruber, L., **Kugler, L.**, Weissmann, M., Dorninger, M., & Serafin, S. (2024). The fractions skill score for ensemble forecast verification. QJRM, 150(764), 4457-4477. <https://doi.org/10.1002/qj.4824>

Kugler L. (2024). Assimilation of cloud-affected satellite observations for convective-scale numerical weather prediction. PhD thesis, University of Vienna. <https://doi.org/10.25365/thesis.75648>

Kugler, L., Anderson, J. L., & Weissmann, M. (2023). Potential impact of all-sky assimilation of visible and infrared satellite observations compared with radar reflectivity for convective-scale numerical weather prediction. QJRM, 149(757), 3623-3644. <https://doi.org/10.1002/qj.4577>

Kugler L. (2019): The Added Value of Machine Learning in Forecasting Wind Turbine Icing. MSc thesis, University of Vienna. <https://doi.org/10.25365/thesis.60595>

CONFERENCE PRESENTATIONS / INVITED TALKS

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|---------|---|
| 06/2026 | 26 th International Conference on Computational Science - Workshop Machine Learning and Data Assimilation for Dynamical Systems, Hamburg, DE
Talk: <i>Scalable Hierarchical Graph Autoencoders for Latent 3D-Variational Assimilation in a Coupled Earth-system Model</i> |
| 05/2026 | Seminar of the Danish Meteorological Institute (online)
Invited Talk: <i>Rethinking Data Assimilation (DA): Can reanalyses improve current variational DA?</i> |
| 04/2025 | General Assembly of the European Geosciences Union, Vienna, AT
Poster: <i>Dealing with unresolved scales of motion and systematic errors in the assimilation of cloud-affected satellite radiances</i> |
| 10/2024 | 10 th International Symposium on Data Assimilation, Kobe, JP
Talk: <i>On the Potential Impact of Visible and Infrared Radiance Assimilation and the Effect of Nonlinear Observation Operators</i> |
| 06/2024 | ISDA seminar on convective-scale data assimilation (online)
Talk: <i>Ambiguity and nonlinearity in the assimilation of visible and infrared observations</i> |

10/2023	9 th International Symposium on Data Assimilation, Bologna, IT Talk: <i>Comparing the assimilation of visible and infrared satellite observations to radar reflectivity for convective-scale numerical weather prediction</i>
05/2023	Meeting of the Austrian Meteorological Society, Innsbruck, AT Poster: <i>The potential impact of assimilating cloud-affected visible and infrared satellite observations for convective-scale numerical weather prediction</i>
04/2023	General Assembly of the European Geosciences Union, Vienna, AT Talk: <i>Assimilating cloud-affected visible & infrared satellite observations in idealized simulations</i>
06/2022	8 th International Symposium on Data Assimilation, Fort Collins, CO, USA Talk: <i>Assimilating visible & infrared satellite observations for convective scale NWP</i>
05/2022	General Assembly of the European Geosciences Union, Vienna, AT Talk: <i>Assimilating visible & infrared satellite observations for convective scale NWP</i>
11/2019	1 st Artificial Intelligence for Copernicus workshop, Reading, UK Invited talk: <i>Machine learning techniques for high-impact-weather (flood forecasts)</i>
09/2019	Workshop of the European Summer of Weather Code (ESoWC), Reading, UK Invited talk: <i>Machine learning techniques for high-impact-weather (flood forecasts)</i>

RESEARCH POSITIONS

12/2025 - 11/2027	Postdoctoral Fellow Faculty of Mathematics and Physics, University of Ljubljana - Fully-coupled Data assimilation for Earth-system models in the latent space discovered by autoencoders - Awarded fellowship in the MSCA COFUND Project "SMASH: Machine learning for Sciences and Humanities"	Ljubljana, SL
10/2025 - 11/2025	Visiting Scientist MMM lab, NCAR Satellite data preprocessing, towards Level-1c satellite data assimilation	Boulder, CO, US
08/2024 - 07/2025	Project Scientist (Postdoctoral Research Scientist) Department of Meteorology and Geophysics, University of Vienna Employed in the project "Demonstrating parameter estimation with ensemble-based DA" (doi: 10.55776/P37259)	Vienna, AT
03/2020 - 04/2024	University Assistant (Doctoral Research Assistant) Department of Meteorology and Geophysics, University of Vienna - Research on the topic "All-sky assimilation of visible and infrared satellite observations for convective-scale numerical weather prediction" - Teaching of Bachelor and Master tutorials (see Teaching section)	Vienna, AT
10/2017 - 08/2019	Assistant Model Developer UBIMET GmbH (private weather service) - Taking care of continuously running forecast models from a software and meteorological perspective - Experiments for improving the used parametrizations and source-code development for the land-surface-model parametrization - Developing data-driven blending methods of forecast models - Rewriting legacy code in python to improve the speed of execution - Providing code to handle large data volumes for the most used file formats in python for plotting, verification, analytics	Vienna, AT

EDUCATION

03/2020 -	PhD Meteorology (Dr.rer.nat)	Vienna, AT
04/2024	Department of Meteorology and Geophysics, University of Vienna Topic: "Data assimilation of cloud-affected satellite observations for convective-scale numerical weather prediction"	
10/2017 -	MSc Meteorology	Vienna, AT
01/2020	Department of Meteorology and Geophysics, University of Vienna Thesis topic: Added Value of Machine Learning in Forecasting Wind Turbine Icing	
10/2013 -	BSc Meteorology	Vienna, AT
09/2017	Department of Meteorology and Geophysics, University of Vienna Thesis topic: Parameterizing the asphalt surface temperature (in German)	

TEACHING EXPERIENCE

Summer 2021 - Summer 2022	"Physical Concepts" , Bachelor Meteorology, University of Vienna The course covered the basics of thermodynamics, radiation and hydrodynamics and was conducted for students of Meteorology and Astronomy as a collaboration between the Department for Meteorology and the Department for Astrophysics
Winter 2022	"Dynamics of the Atmosphere" , Bachelor Meteorology, University of Vienna The course covered fundamental forces, equations of motion in various coordinates, approximations to the equations, vorticity equation, atmospheric waves.
Summer 2023	"Advanced Data Assimilation" , Master Meteorology, University of Vienna I pioneered a hands-on course where students could learn concepts of data assimilation with practical examples using the Data Assimilation Research Testbed (DART) and forecast data from the Weather Research and Forecasting Model (WRF). The course introduced DART, WRF and covered Bayes theorem, forecast verification, sequential assimilation, localization, inflation, overfitting of observations and the theoretical optimum for spread.

PROGRAMMING

- **Programming languages, etc:** Python, Fortran, Bash, Matlab, Git, LaTeX
- **HPC knowledge/skills** from various HPC systems; for speed-up and efficiency of code using MPI (python/Fortran) and OpenMP; learned through experience, semester-long elective courses "Computer Architecture and High Performance Computing", "Databases and Visualization", "Algorithms and Data Structures", and PRACE HPC workshops in Barcelona and Amsterdam (2018)
- **Open-source software projects:**
 - [DART-WRF](#): Digital lab for observing-system simulation experiments with WRF, DART, and RTTOV to quickly iterate through a large number of various configurations, managing concurrently running experiments and program dependencies through HPC queueing systems.
 - [RTTOV-WRF-MPAS](#): Simulate SEVIRI satellite observations from WRF/MPAS NWP model output.
 - [ECMWFCode4Earth/ml_flood](#): Demonstrator and tutorial for ML in river flood forecasting and other projects
- **Contributions to other software projects:**
 - [NCAR/DART](#): Development, testing and maintenance of the RTTOV radiative transfer interface

OTHER

- **Languages:** German (native), English
- **Citizenship:** Austrian